

# Choose an efficient calculation method

Use the scaffold, show each step and ask for an adult check when marked.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Read the model: Choose an efficient method for  $3,998 + 2,507$  and calculate. Copy or complete the first step.

2. Choose an efficient method for  $6,300 \div 25$ .

3. Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve.

4. Miro says: Miro believes the longest written method is always the safest choice. Is this correct? Explain.

# Teacher answers and checking prompts

1. Read the model: Choose an efficient method for  $3,998 + 2,507$  and calculate. Copy or complete the first step.

**Answer** Compensation:  $4,000 + 2,507 - 2 = 6,505$ .

2. Choose an efficient method for  $6,300 \div 25$ .

**Answer** Multiply by 4 then divide by 100, or use known multiples: 252.

3. Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve.

**Answer** A written method is efficient; the product is 339,572.

4. Miro says: Miro believes the longest written method is always the safest choice. Is this correct? Explain.

**Answer** An efficient method fits the numbers and still allows a reliable check.

# Choose an efficient calculation method

Work independently. Show a complete method and check at least one answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Choose an efficient method for  $3,998 + 2,507$  and calculate.

6. Check this result using an inverse, estimate or known fact: Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve.

2. Choose an efficient method for  $6,300 / 25$ .

7. Create one calculation best solved mentally and one best solved using a written method. Justify both choices.

3. Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve.

8. Identify and correct this misconception: Miro believes the longest written method is always the safest choice.

4. Solve this independently and show every stage: Choose an efficient method for  $3,998 + 2,507$  and calculate.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Choose an efficient method for  $6,300 / 25$ .

10. Write and solve a similar question to: Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve.

# Teacher answers and checking prompts

1. Choose an efficient method for  $3,998 + 2,507$  and calculate.  
**Answer** Compensation:  $4,000 + 2,507 - 2 = 6,505$ .
2. Choose an efficient method for  $6,300 \div 25$ .  
**Answer** Multiply by 4 then divide by 100, or use known multiples: 252.
3. Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve.  
**Answer** A written method is efficient; the product is 339,572.
4. Solve this independently and show every stage: Choose an efficient method for  $3,998 + 2,507$  and calculate.  
**Answer** Compensation:  $4,000 + 2,507 - 2 = 6,505$ .
5. Use a second method or representation for: Choose an efficient method for  $6,300 \div 25$ .  
**Answer** Expected result: Multiply by 4 then divide by 100, or use known multiples: 252. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve.  
**Answer** Expected result: A written method is efficient; the product is 339,572. A valid check must be shown.
7. Create one calculation best solved mentally and one best solved using a written method. Justify both choices.  
**Answer** Answers vary. The justification should refer to number structure and accuracy.
8. Identify and correct this misconception: Miro believes the longest written method is always the safest choice.  
**Answer** An efficient method fits the numbers and still allows a reliable check.
9. Write a complete sentence explaining how you checked one answer.  
**Answer** Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve.  
**Answer** Answers vary. The new question must use the same mathematical structure and include a correct solution.

# Choose an efficient calculation method

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Create one calculation best solved mentally and one best solved using a written method. Justify both choices.

6. Create a new example that tests the same idea as: Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve.

2. Prove the answer to this question, rather than only calculating it: Choose an efficient method for  $3,998 + 2,507$  and calculate.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Choose an efficient method for  $6,300 \div 25$ .

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: A written method is efficient; the product is 339,572.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro believes the longest written method is always the safest choice.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

# Teacher answers and checking prompts

1. Create one calculation best solved mentally and one best solved using a written method. Justify both choices.  
**Answer** Answers vary. The justification should refer to number structure and accuracy.
2. Prove the answer to this question, rather than only calculating it: Choose an efficient method for  $3,998 + 2,507$  and calculate.  
**Answer** Expected result: Compensation:  $4,000 + 2,507 - 2 = 6,505$ . Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Choose an efficient method for  $6,300 \div 25$ .  
**Answer** Expected result: Multiply by 4 then divide by 100, or use known multiples: 252. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: A written method is efficient; the product is 339,572.  
**Answer** One valid reconstruction is: Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro believes the longest written method is always the safest choice.  
**Answer** An efficient method fits the numbers and still allows a reliable check.
6. Create a new example that tests the same idea as: Would you use mental or written calculation for  $7,382 \times 46$ ? Explain and solve.  
**Answer** Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.  
**Answer** Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.  
**Answer** Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.  
**Answer** Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.  
**Answer** Answers vary. The explanation must distinguish the mathematical structure from the changed value.